

Out-of-Pocket Healthcare Costs and Poverty Analysis

1. PROJECT OVERVIEW & CONTEXT

This research provides a rigorous statistical analysis of primary data mapping out-of-pocket healthcare expenditures (OOPE) across **300 surveyed households** in Kolhapur city. The primary goal is to isolate medical and non-medical cost blocks, track risk disparities among inpatient vs. outpatient metrics, and quantify health-induced poverty dynamics using advanced inferential modeling (Factorial ANOVA, Ridge Regression, MANOVA, and Logit Models).

300

TOTAL SAMPLES

₹2,611.30

MEAN TOTAL EXP

9.43%

MEAN BURDEN RATIO

₹13,000

MAX RECORDED COST

2. DESCRIPTIVE ANALYSIS OF HEALTHCARE SPENDING

The total expenditure structure is heavily split between medical cost items (doctor consultation fees, hospitalization paper costs, drugs, and tests) and secondary non-medical items (transit tickets and companion food setups).

STATISTICAL PROPERTY	MEDICAL COST (₹)	NON-MEDICAL COST (₹)	TOTAL OOP EXPENDITURE (₹)	HEALTH BURDEN RATIO
Mean (Average)	2,395.24	216.30	2,611.30	0.0943 (9.43%)
Standard Deviation	2,452.05	224.50	2,621.81	0.1108
Median (50th Percentile)	1,100.50	51.00	1,401.00	0.0500 (5.00%)
75th Percentile	3,601.00	300.50	3,901.50	0.1400 (14.00%)

3. SOCIO-ECONOMIC STRATIFICATION & POVERTY STATUS

Poverty impact benchmarks were calculated based on post-expenditure household budget reductions. Households facing inpatient treatments or private hospital admissions suffer significantly higher ratios of financial distress.

STRATIFICATION FACTOR	POOR RATIO (%)	NON-POOR RATIO (%)
Inpatient Admissions	25.41%	74.59%
Outpatient Care	17.39%	82.61%
Government Hospital Users	19.86%	80.14%
Private Hospital Users	24.67%	75.33%
Urban Residence Cohort	22.35%	77.65%
Rural Residence Cohort	22.31%	77.69%

4. ADVANCED INFERENTIAL MODELING FINDINGS

A. Ridge Regression Analysis (Cost Drivers)

A multi-variable Ridge regression framework was mapped to isolate the heaviest cost drivers contributing to the overall financial shock. ****Medicine costs**** and ****diagnostic testing costs**** showed the highest coefficients, highlighting them as major components of outpatient expenses.

- **Medicine Cost Coefficient:** +1,420.58
- **Diagnostics Cost Coefficient:** +1,147.57
- **Hospital Visit Cost Coefficient:** +280.72
- **Consultation Fee Coefficient:** +216.78

B. Factorial ANOVA & MANOVA (Intergroup Variations)

A Factorial ANOVA was run to track total spending variance across family groupings, geographic setups, and insurance status. In parallel, a Multivariate ANOVA (MANOVA) verified the joint impact on total health expenditure and poverty status.

FACTOR CATEGORY	ANOVA F-STATISTIC	ANOVA P-VALUE	MANOVA WILKS' LAMBDA	STATISTICAL RESULT
Family Type (Joint vs Nuclear)	16.7602	0.000055	0.9410	Highly Significant
Insurance Enrollment Status	16.2116	0.000072	0.8931	Highly Significant
Residence Area (Urban/Rural)	9.6597	0.002100	0.9657	Significant

C. Binary Logistic Regression (Poverty Odds Model)

A binary logistic regression model was constructed to predict the probability of a household falling into poverty due to medical bills. Nuclear families and families enrolled in health insurance schemes exhibited significantly lower risks of health-induced financial distress.

- **Family Type (Nuclear Structure):** Coeff: -1.2835 ($z = -3.634$, $p = 0.000$) → Reduces poverty risk
- **Insurance Enrolled (Yes):** Coeff: -1.0690 ($z = -3.021$, $p = 0.003$) → Provides major financial safety netting
- **Residence Area (Urban):** Coeff: +0.4745 ($z = 1.498$, $p = 0.134$) → Not statistically significant.

5. INSURANCE UTILIZATION GAP & CHI-SQUARE TEST

Chi-Square Test of Association

A chi-square test was executed to examine whether enrollment in an insurance program translates directly into successful claims usage during treatments. With a **Chi-square statistic of 73.6607** and an extremely low **p-value of 9.28×10^{-18}** , the hypothesis of independence was soundly rejected.

This points to a significant structural gap: while enrollment is high, actual claims usage is limited by processing delays or complex procedures in private hospitals.

6. STRATEGIC POLICY RECOMMENDATIONS

- **Regulate Outpatient Pharmacy Pricing:** Since medicine costs are the leading driver of financial distress (+1,420.58), expanding access to generic medicine centers (Pradhan Mantri Bhartiya Janaushadhi Pariyojana) across Kolhapur is highly recommended.
- **Optimize Claims Processing Layouts:** Simplify the process for utilizing government schemes (like the Mahatma Jyotirao Phule Jan Arogya Yojana) in private hospitals to address the high financial risk (24.67% poverty ratio) observed in private facilities.